

Why did throughput drop on our holiday shift?

Flow rate fell noticeably on Company Founding Day. Was it mechanical — or simply staffing?

—36%

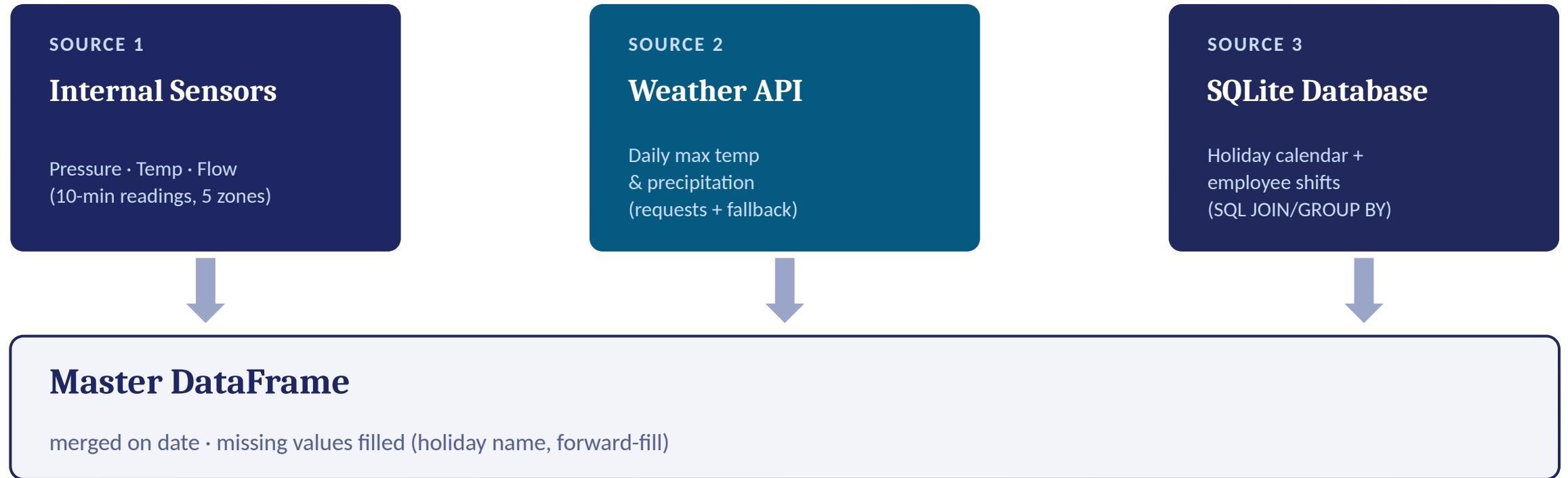
avg. flow rate on the holiday vs. the weekly average

50%

of normal staff scheduled that day

One week of sensor data raised the question — this analysis pulls in weather and staffing data to answer it.

Three sources, one daily timeline



THE INSIGHT & RECOMMENDATION

Staffing explains the dip — not the equipment

The merged dataset shows total scheduled staff correlates with daily flow rate ($r \approx 0.4$ direction; the holiday, at half staffing, has both the lowest headcount and the lowest throughput of the week).

RECOMMENDATION

Model minimum viable staffing for known holidays before next year's schedule ships, and confirm with a maintenance check that no mechanical issue is masking the staffing effect.

0.40

correlation: staffing vs.
average flow rate

1 wk

of data analyzed —
next step: re-check over a full month

3

data sources unified:
sensors + weather + HR system